UHR-DETR: Efficient End-to-End Small Object Detection for Ultrahigh-Resolution Remote Sensing Imagery

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***Abstract*—The ultrahigh-resolution (UHR) imagery has become essential for modern remote sensing, offering unprecedented spatial coverage. However, detecting small objects in such vast scenes presents a critical dilemma: retaining the original resolution for small objects causes prohibitive memory bottlenecks. Conversely, the conventional compromises, such as image downsampling or patch cropping, either erase small objects or destroy context. To break this dilemma, we propose UHR-DETR, an efficient end-to-end transformer-based detector designed for UHR imagery. First, we introduce a coverage-maximizing sparse encoder that dynamically allocates finite computational resources to informative high-resolution regions, ensuring maximum object coverage with minimal spatial redundancy. Second, we design a global–local decoupled decoder. By integrating macroscopic scene awareness with microscopic object details, this module resolves semantic ambiguities and prevents scene fragmentation. Extensive experiments on the UHR imagery datasets (e.g., STAR and SODA-A) demonstrate the superiority of UHR-DETR under strict hardware constraints (e.g., a single 24GB RTX 3090). It achieves a 2.8% mean average precision (mAP) improvement while delivering a 9× inference speedup compared to standard sliding-window baselines on the STAR dataset. Our code and models are available at https://github.com/Li-JingFang/UHR-DETR**

***Index Terms*—Detection transformer, small object detection, ultrahigh-resolution (UHR) remote sensing imagery.**

I. Introduction

DETECTING small objects in remote sensing imagery has achieved remarkable progress [1], [2], serving as a cornerstone for macroscopic applications, such as urban planning and maritime surveillance. However, driven by the rapid development of modern Earth observation systems and the growing demand for continuous wide-area monitoring, ultrahigh-resolution (UHR) imagery is rapidly establishing itself as the new standard for macroscopic perception [3]. For instance, given a standard ground sample distance (GSD) of 0.5 m/pixel, DOTA benchmark [4] images (e.g., 1024 × 1024) cover a highly restricted physical area of merely 0.26 km^2. In stark contrast, a UHR image easily spans 8192 × 8192 pixels [5], encompassing an extensive area of nearly 17 km^2. This leap in spatial dimensions imposes unprecedented demands on object detectors. The challenge is further exacerbated by the fact that perceiving small objects requires preserving fine-grained, high-resolution features to handle their densely packed distributions and the extreme scale disparity against the vast UHR canvas. Consequently, maintaining these high-resolution representations across such expansive spatial extents substantially increases memory consumption and creates severe bottlenecks in inference speed.

To alleviate the computational burden of UHR imagery, the existing UHR object detection strategies generally fall into two categories. The first approach utilizes efficient nonuniform downsampling strategies to preserve a larger pixel ratio for foreground regions [6], [7], [8]. Alternatively, the second approach introduces selective filtering mechanisms into the standard sliding-window framework to skip empty backgrounds and improve detection efficiency [9], [10]. Yet, despite the progress made over naive downsampling and exhaustive cropping, detecting small objects in massive scenes still faces profound challenges. The first challenge is the ***severe conflict between resolution and efficiency***: precise detection of small objects (under 32 × 32 pixels [11]) inherently requires high-resolution features, making it incredibly difficult to efficiently extract such features from UHR images (over 8192 × 8192 pixels) without exhausting computational resources. As shown in Fig. 1(b), increasing the downsampling ratio results in a substantial 96% relative reduction in small-object accuracy. Second, the robust perception of small objects heavily relies on sufficient contextual support to alleviate semantic ambiguity. Since sliding window strategies inevitably ***fracture the global image topology***, effectively reconstructing and bridging this severed macroscopic context to aid microscopic detection remains a critical unsolved obstacle. As shown in Fig. 1(c), rigid sliding windows isolate microscopic details (e.g., solar panels) from their corresponding macroscopic context (e.g., the power plant), depriving the detector of crucial semantic cues; quantitatively, Table I shows that restricting detection

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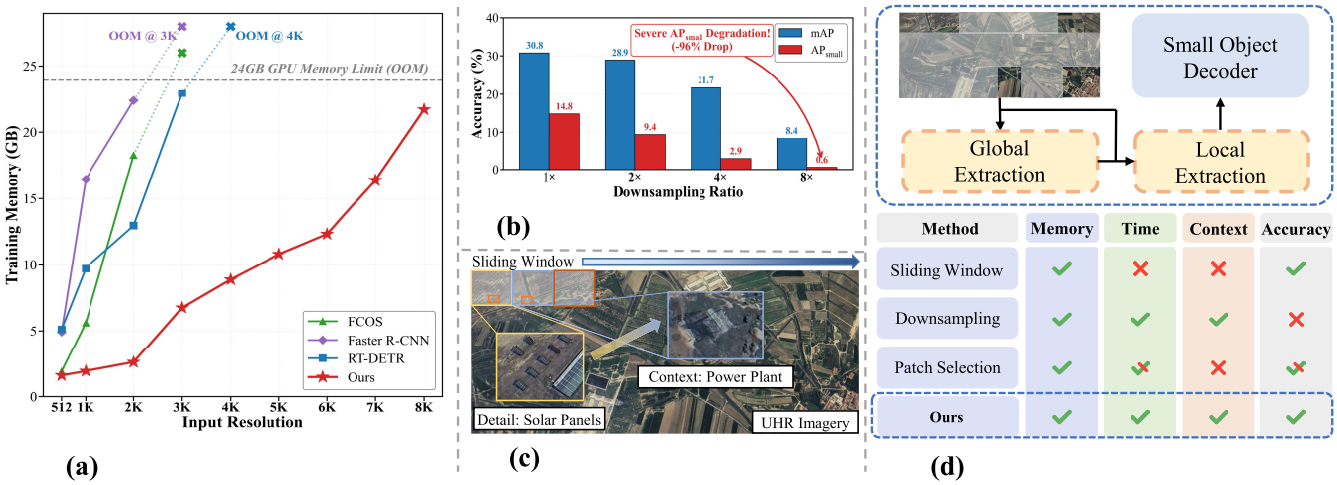


Fig. 1. Perception paradox in UHR imagery and our solution. (a) Standard object detectors suffer from severe out-of-memory (OOM) bottlenecks on 24-GB GPUs, whereas our method efficiently scales up to ultrahigh resolutions. (b) Naive downsampling leads to a catastrophic drop in small object accuracy. (c) Sliding window strategies inevitably fracture crucial macroscopic context. (d) To break these dilemmas, UHR-DETR introduces a global–local decoupled paradigm, achieving an optimal balance among memory efficiency, inference speed, context preservation, and detection accuracy.

TABLE I

EFFECT OF WINDOW-BASED INFERENCE ON DETECTION PERFORMANCE. ALL VARIANTS USE FCOS AND ARE EVALUATED ON THE SAME 2048 × 2048 IMAGES. "FULL IMAGE" DENOTES TRAINING AND INFERENCE ON THE ENTIRE IMAGE, WHEREAS THE SLIDING-WINDOW VARIANTS ARE TRAINED ON 512 × 512 CROPS AND PERFORM INFERENCE USING 512 × 512 WINDOWS

Inference Scheme	Overlap	mAP	AP_S	AP_M	AP_L
Sliding window	0	31.3	12.5	34.3	34.0
Sliding window	128	33.1	13.2	35.8	37.2
Full image	–	40.2	17.0	42.1	44.3

to local windows leads to a substantial accuracy degradation, which cannot be fully recovered by introducing window overlap.

Motivated by these challenges, and building upon the advanced RT-DETR [12] architecture, we propose UHR-DETR, the first efficient, end-to-end transformer-based detector designed for UHR imagery. As illustrated in Fig. 1(a), the UHR-DETR bypasses the prohibitive memory and latency bottlenecks that paralyze standard models processing massive inputs. Moreover, it resolves the two aforementioned dilemmas through two core innovations. First, to tackle the severe conflict between resolution and efficiency, we draw inspiration from the "glimpse-and-fixate" mechanism of the human visual system. Just as humans naturally leverage a macroscopic glimpse to direct high-resolution foveal attention exclusively toward critical localized regions, we introduce a coverage-maximizing sparse encoder to mimic this biological paradigm. Instead of exhaustively processing the entire image, this module leverages a lightweight backbone to extract global features and formulates spatial feature routing as a geometric set cover problem. Optimized via a novel local peak margin loss and executed through an iterative soft-subtraction greedy algorithm (ISSGA), it dynamically allocates finite computational resources strictly to informative high-resolution regions. By doing so, the encoder effectively extracts the crucial regional features containing objects of interest, significantly reducing spatial redundancy by bypassing vast uninformative backgrounds. Second, to reconstruct the severed holistic context, we design a global–local decoupled decoder. To circumvent the compromises of conventional patch inference and aggressive downsampling, our decoder introduces a progressive global-to-local dual cross-attention paradigm. The object queries initially interact with global features to establish macroscopic scene awareness and are subsequently routed to the sparse high-resolution patches to perceive microscopic object details. This decoupled design explicitly integrates global topology with local refinement, facilitating robust local feature discrimination through global contextual guidance. Extensive experiments on two massive benchmarks, STAR [3] (8192 × 8192) and SODA-A [13] (9600 × 9600), demonstrate the superiority of UHR-DETR in handling densely packed small objects. Notably, our method achieves a 2.8 mean average precision (mAP) improvement and over a 9× inference speedup compared to the exhaustive sliding-window RT-DETR. Moreover, it consistently surpasses the existing state-of-the-art UHR detectors by a significant margin of 4.2 mAP, all while maintaining faster processing speeds.

Our main contributions are summarized as follows.

1) We propose UHR-DETR, the first efficient, end-to-end transformer-based detector specifically designed for small object detection in UHR remote sensing imagery. It addresses severe memory and computational bottlenecks, providing a holistic alternative to the traditional fragmented crop-and-detect paradigm.
2) We introduce a detection architecture featuring a coverage-maximizing sparse encoder and a global–local decoupled decoder. This design dynamically routes finite computation to high-value regions via a geometric set

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cover formulation and seamlessly fuses macroscopic scene context with microscopic details.

3) Extensive experiments on the STAR and SODA-A datasets validate the effectiveness of UHR-DETR. Compared to standard baselines, our method maintains comparable detection accuracy while achieving over a 9× reduction in inference latency, successfully processing massive UHR inputs and overcoming the severe memory limitations inherent in standard architectures.

The remainder of this article is organized as follows. First, we review the related work on transformer-based small object detection and UHR object detection in Section II. Next, we detail the proposed UHR-DETR framework, focusing on the coverage-maximizing sparse encoder and the global–local decoupled decoder, in Section III. Subsequently, we present extensive experiments and comprehensive ablation studies conducted on multiple public datasets in Section IV. Additionally, we discuss the underlying design principles and inherent limitations of our approach in Section V. Finally, we conclude this article in Section VI.

II. Related Work

A. Transformer-Based Small Object Detection

Detecting small objects has been a central focus in the remote sensing community. Owing to their ability to model long-range contextual dependencies, transformer-based architectures have become the dominant paradigm for small object detection. The existing transformer-based detection methods can be broadly categorized into two directions: feature representation enhancement and learning strategies optimization.

Feature representation enhancement aims to leverage transformers' attention mechanisms to assign more discriminative features to small objects. Deformable DETR [14] first introduced deformable attention, which combines the sparse sampling capability of DCN with the global relational modeling of transformers, significantly enhancing the representation of small objects over DETR. DART [15] further improves spatial perception for small objects by dynamically generating content-aware regions via an adaptive-region attention mechanism. Cross-DINO [16] incorporates a deep MLP to aggregate initial features with both short- and long-range information, strengthening representations for small objects. D^3R-DETR [17] exploits the frequency-domain features to better capture contextual cues for small objects. Learning strategy optimization focuses on improving transformer queries and Hungarian matching to enable faster querying and better convergence for small objects. Co-DETR [18] introduces an auxiliary decoder that leverages dense label assignment (e.g., FCOS [19], ATSS [20]) to enhance convergence. DQ-DETR [21], D3Q-DETR [22], Dome-DETR [23], and DEAL [24] generate query numbers from density maps (via classification or regression) and dynamically allocate queries during training, significantly improving performance on dense small object detection.

Although transformer-based methods for small object detection have achieved notable progress, their reliance on global contextual modeling leads to substantial memory consumption as image resolution increases. To address this limitation, we propose a coverage-maximizing sparse encoder that leverages sparse modeling to improve efficiency without sacrificing performance.

B. UHR Object Detection

Driven by rapid advancements in sensor technologies, the UHR imagery presents unique challenges for small object detection due to the extreme scale variation between vast backgrounds and densely packed small objects.

To systematically benchmark algorithms in this domain, the evaluation datasets have progressively expanded in spatial coverage. Early efforts primarily focused on standard high-resolution drone imagery, represented by datasets, such as VisDrone [25] and UAVDT [26]. However, to address wide-area perception, the community has transitioned to massive benchmarks, such as PANDA [27] for gigapixel crowds and STAR [3] for holistic satellite perception. From an algorithmic perspective, various strategies have been proposed to alleviate the computational burden on these massive images. Macrolevel approaches selectively process informative regions by leveraging clustering algorithms [9], [10], [28], [29] and density maps [30], [31], [32] to isolate patches rich in small objects or by utilizing nonuniform sampling [6], [7], [8], [33] to elastically compress backgrounds. Alternatively, micro-level architectures attempt to optimize efficiency by dynamically routing computation exclusively to informative tokens [34], [35], [36].

Although these efficiency-oriented designs substantially reduce computational overhead, they often sacrifice contextual information in UHR imagery, which is crucial for the semantic understanding of small objects. To address this issue, we propose a global–local decoupled decoder that reintroduces global context at the decoding stage, thereby enhancing the representation of small objects.

III. Approach

A. Overall Architecture

The overall architecture of the proposed UHR-DETR, illustrated in Fig. 2, is tailored for efficient object detection in UHR imagery. Unlike the existing DETR-style frameworks that primarily target object detection under conventional resolutions, UHR-DETR is specifically designed to address the unique computational and contextual challenges posed by UHR inputs. The framework consists of two key components: **coverage-maximizing sparse encoder** and **global–local decoupled decoder**. Coverage-maximizing sparse encoder aims to efficiently encode UHR images by sparsely sampling informative regions under global guidance, thereby reducing computational overhead while preserving essential contextual coverage. Global–local decoupled decoder is designed to integrate global scene context with fine-grained local features during decoding, enabling accurate localization and recognition of small objects.

Formally, given a massive UHR input image $\mathcal{I} \in \mathbb{R}^{H\times W\times 3}$, it is first downsampled via a max pooling layer. Then, a lightweight feature extractor (e.g., ResNet-18 [37]), denoted

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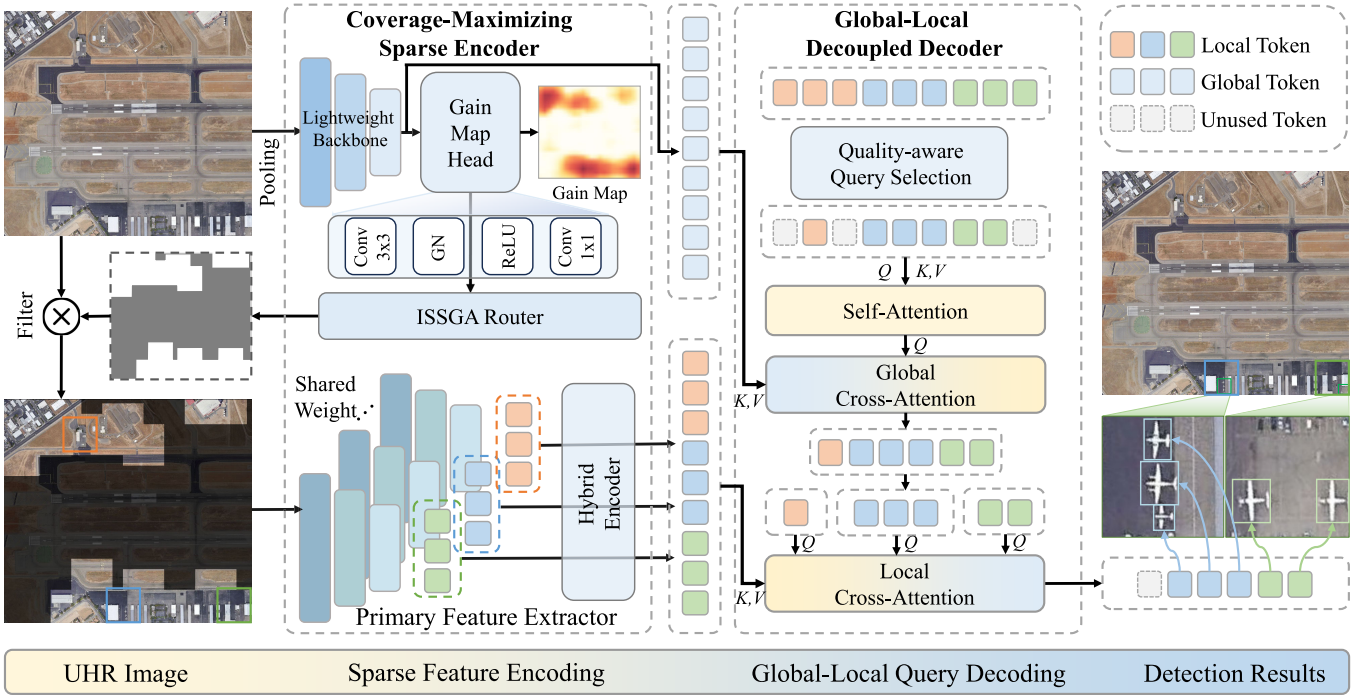


Fig. 2. Overall architecture of the proposed UHR-DETR. The framework comprises two modules. (Left) Coverage-maximizing sparse encoder utilizes a lightweight backbone to generate global tokens and an object-aware gain map. Guided by this map, an ISSGA dynamically extracts high-value image patches, which are subsequently processed by the primary feature extractor to construct fine-grained local tokens. (Right) Global–local decoupled decoder hierarchically refines object queries. It sequentially performs GCA to inject macroscopic spatial priors, followed by local cross-attention to accurately perceive microscopic details, yielding a unified and highly efficient end-to-end UHR perception pipeline.

as Φ_{light}, processes this downsampled input to extract global tokens $\mathcal{F}_{\text{global}} \in \mathbb{R}^{h\times w\times c}$, capturing the macroscopic topology of the scene

$$\mathcal{F}_{\text{global}} = \Phi_{\text{light}}\left(\text{MaxPool}\left(\mathcal{I}\right)\right). \tag{1}$$

A gain map head subsequently acts on $\mathcal{F}_{\text{global}}$ to predict an object-aware gain map, which explicitly quantifies the potential object coverage of any given subregion. Guided by this map, an iterative greedy algorithm strategically crops K (default $K = 40$) high-value image patches $\{\mathcal{P}_k\}_{k=1}^{K}$ from the original high-resolution image $\mathcal{I}$. These sparse but highly informative local regions are then fed into the primary feature extractor (we use the backbone and hybrid encoder of RT-DETR [12] as our default feature extractor), parameterized as Φ_p, to construct the high-resolution local tokens $\mathcal{F}_{\text{local}}$

$$\mathcal{F}_{\text{local}} = \bigcup_{k=1}^{K} \Phi_p\left(\mathcal{P}_k\right). \tag{2}$$

Finally, a set of object queries are processed through the global–local decoupled decoder. By sequentially performing global cross-attention (GCA) with $\mathcal{F}_{\text{global}}$ and local cross-attention with $\mathcal{F}_{\text{local}}$, the queries seamlessly integrate macroscopic scene awareness with microscopic object details, ultimately predicting accurate bounding boxes in a unified global coordinate system.

B. Coverage-Maximizing Sparse Encoder

Processing the entire UHR image at its original resolution incurs prohibitive computational costs and severe memory bottlenecks, especially given that small objects are typically sparsely distributed across vast empty backgrounds. To resolve this inefficiency, the coverage-maximizing sparse encoder is designed to dynamically allocate finite computational resources to highly informative regions. To achieve this, we evaluate the perceptual value of each candidate subregion across the vast UHR canvas. Guided by these regional value estimates, we then formulate the selective spatial routing as a geometric set cover problem, aiming to maximize the inclusion of small objects while minimizing spatial redundancy. Specifically, this module operates through a systematic two-step pipeline: it first performs *global object distribution estimation* by predicting a dense gain map to quantify macroscopic object density and subsequently executes *dynamic sparse patch routing* via a tailored iterative greedy algorithm to strategically sample the optimal subset of high-resolution patches.

1) Global Object Distribution Estimation: To explicitly identify high-value regions without processing the entire UHR image at full resolution, this module is designed to estimate the spatial distribution of small objects. Taking the global tokens $\mathcal{F}_{\text{global}} \in \mathbb{R}^{h\times w\times c}$ as input, it produces a spatial gain map. The spatial dimensions $h \times w$ naturally divide the original UHR canvas into a dense grid of candidate patches. Each location (x, y) on the gain map outputs a value score, representing the expected object coverage within a high-resolution patch of size $p_H \times p_W$ (set to 512×512 by default) centered at that coordinate. Specifically, for an input of size $H \times W$ and a gain map of size $h \times w$, location (x, y) is mapped to the image-space center $((x + 0.5)W/w, (y + 0.5)H/h)$. On STAR, each

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cell of the 64 × 64 gain map corresponds to a stride of 128 pixels in the 8192 × 8192 input, so a 512 × 512 patch spans 4 × 4 gain map cells.

To robustly estimate the object quantity within each patch, we recast this unbounded dense regression into a classification-based expectation task. The gain map head projects the input features $\mathcal{F}_{\text{global}}$ into a logit tensor $Z \in \mathbb{R}^{h\times w\times(M+1)}$, where M is a predefined hyperparameter representing the discrete bin limit. Let $Z_{x,y,b}$ denote the predicted logit for the discrete bin $b \in \{0, 1, \ldots, M\}$ at spatial location (x, y). The final continuous predicted score, denoted by $\mathcal{G}_{(x,y)}$, is computed by mapping the discrete probability distribution to the continuous coverage scale [38]

$$\mathcal{G}_{(x,y)} = \sum_{b=0}^{M} b^2 \cdot \frac{\exp\left(Z_{x,y,b}\right)}{\sum_{j=0}^{M} \exp\left(Z_{x,y,j}\right)}. \tag{3}$$

During training, this predicted score is supervised by the actual object coverage. The bounded ground-truth target, denoted as $\mathcal{G}^{GT}_{(x,y)}$, is defined based on the intersection over foreground (IoF). To prevent regions with extremely dense object clusters from dominating the learning gradients and to ensure optimization stability, we directly cap the raw IoF sum at a maximum threshold of M^2

$$\mathcal{G}^{GT}_{(x,y)} = \min\left(\sum_{i=1}^{N} \frac{\text{Area}\left(\mathcal{P}_{(x,y)} \cap B_i\right)}{\text{Area}\left(B_i\right)}, M^2\right) \tag{4}$$

where Area(B_i) denotes the spatial area of the ith ground-truth bounding box. We then apply the distribution focal loss [38] to supervise the discrete probability distribution against this continuous target $\mathcal{G}^{GT}_{(x,y)}$.

Furthermore, to explicitly sharpen the predicted local maxima around object clusters, we design an auxiliary objective termed the **local peak margin loss**. Denoting Ω as the set of ground-truth peak locations and Δp as the 4-neighborhood shifts, the loss $\mathcal{L}_{LPM}$ penalizes any neighboring prediction $\mathcal{G}_{p+\Delta p}$ that encroaches within a predefined margin Δ of the peak value $\mathcal{G}_p$

$$\mathcal{L}_{LPM} = \frac{1}{|\Omega|} \sum_{p\in\Omega} \max\left(0, \mathcal{G}_{p+\Delta p} + \Delta - \mathcal{G}_p\right). \tag{5}$$

2) Dynamic Sparse Patch Routing: Taking the predicted gain map $\mathcal{G}$ as an input, we select K high-resolution patches to maximize object coverage while reducing spatial redundancy. Let $\mathcal{U}$ denote the set of objects and $S_{(x,y)} \subseteq \mathcal{U}$ the objects covered by a patch centered at (x, y). The routing objective is formulated as

$$\max_{\mathcal{C}} \left| \bigcup_{(x,y)\in\mathcal{C}} S_{(x,y)} \right| \quad \text{s.t.} \quad |\mathcal{C}| = K. \tag{6}$$

Since exact maximum coverage is NP-hard, we adopt a greedy approximation that iteratively selects the location with the highest predicted gain.

Unlike rigid suppression, which completely excludes overlapping candidates, we softly discount their gains to preserve necessary intersections between patches covering dense object clusters. For an offset $(\Delta x, \Delta y)$ from the selected location, the overlap penalty is modeled by a 2-D linear decay kernel

$$\mathcal{K}\left(\Delta x, \Delta y\right) = \left(1 - \frac{|\Delta x|}{p_w}\right)^{+} \left(1 - \frac{|\Delta y|}{p_h}\right)^{+} \tag{7}$$

where p_w and p_h denote the patch dimensions projected onto the gain map. The resulting greedy routing procedure is summarized in Algorithm 1. Upon selecting the peak (x_k, y_k) with $v_k = \mathcal{G}_{(x_k,y_k)}$, the gains within its overlapping neighborhood are softly discounted. Specifically, for each $(x, y) \in \mathcal{N}_{p_w,p_h}(x_k, y_k)$

$$\mathcal{G}_{(x,y)} \leftarrow \left(\mathcal{G}_{(x,y)} - v_k \cdot \mathcal{K}\left(x - x_k, y - y_k\right)\right)^{+} \tag{8}$$

where $\mathcal{N}_{p_w,p_h}(x_k, y_k)$ denotes the local neighborhood centered at (x_k, y_k) and determined by the patch size $p_w \times p_h$, and $(z)^+ = \max(z, 0)$. Locations outside this neighborhood remain unchanged. Since $\mathcal{K}(0, 0) = 1$, the selected peak is removed, while neighboring gains are discounted according to their overlap with the selected patch.

This iterative suppression explicitly penalizes severe overlaps while permitting slight, necessary intersections, ensuring the selected K patches are both diverse and densely packed around target clusters. The cropped patches are then processed by the primary encoder to construct the fine-grained $\mathcal{F}_{\text{local}}$.

Algorithm 1 Iterative Soft-Subtraction Greedy Algorithm

Require: Gain Map $\mathcal{G} \in \mathbb{R}^{h\times w}$, Patch Size $p_h \times p_w$, Budget K
Ensure: Coordinates $\mathcal{C}$, Scores $\mathcal{S}$

$\mathcal{C} \leftarrow \emptyset$; $\mathcal{S} \leftarrow \emptyset$
▷ *Build linear decay overlap kernel*
for $\Delta x \in [-p_w, p_w]$, $\Delta y \in [-p_h, p_h]$ **do**
  $\mathcal{K}(\Delta x, \Delta y) \leftarrow \left(1 - \frac{|\Delta x|}{p_w}\right)^{+} \cdot \left(1 - \frac{|\Delta y|}{p_h}\right)^{+}$
end for
▷ *Greedy selection with soft suppression*
for $k = 1$ **to** K **do**
  $(x_k, y_k) \leftarrow \arg\max_{x,y} \mathcal{G}_{(x,y)}$
  $v_k \leftarrow \mathcal{G}_{(x_k,y_k)}$
  $\mathcal{C} \leftarrow \mathcal{C} \cup \{(x_k, y_k)\}$; $\mathcal{S} \leftarrow \mathcal{S} \cup \{v_k\}$
  if $v_k > 0$ **then**
    for each $(x, y) \in \mathcal{N}_{p_w,p_h}(x_k, y_k)$ **do**
      $\mathcal{G}_{(x,y)} \leftarrow \left(\mathcal{G}_{(x,y)} - v_k \cdot \mathcal{K}(x - x_k, y - y_k)\right)^{+}$
    end for
    $\mathcal{G}_{(x_k,y_k)} \leftarrow 0$
  end if
end for
return $\mathcal{C}$, $\mathcal{S}$

$(z)^+ = \max(z, 0)$; $\mathcal{N}_{p_w,p_h}(x_k, y_k) = \{(x, y) : |x - x_k| \le p_w, |y - y_k| \le p_h\} \cap [0, w) \times [0, h)$.

C. Global–Local Decoupled Decoder

To resolve the severe context loss inherent in isolated patch inference, our global–local decoupled decoder divides the semantic interaction into two hierarchical stages: *global macrointeraction* and *local microinteraction*. This decoupled design seamlessly bridges macroscopic scene awareness with microscopic object details.

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Leveraging the macroscopic global tokens $\mathcal{F}_{\text{global}}$ and the microscopic local tokens $\mathcal{F}_{\text{local}}$ generated by the coverage-maximizing sparse encoder, we first initialize a set of object queries Q. The number of Q is dynamically determined proportional to the spatial sum of the predicted gain map, implicitly adapting to the actual scene density. Specifically, each decoder layer is structured as a systematic three-step interaction pipeline: it begins with a self-attention module to capture interobject dependencies, followed by a GCA module to mine the correlations between objects and the global context, and concludes with a local cross-attention module to associate the objects with local fine-grained features.

Global Macrointeraction: To avoid being blindly trapped in isolated patches and to acquire essential holistic context, the object queries perform multihead cross-attention (MHCA) [39] with the global tokens $\mathcal{F}_{\text{global}}$. Following DAB-DETR [40], we formulate the queries as dynamic anchor boxes to explicitly parameterize their spatial priors. Let $A = (cx, cy, w, h) \in [0, 1]^4$ denote the queries' anchor boxes in the absolute normalized coordinate space. By applying a sinusoidal positional encoding function PE(·), the anchor boxes are spatially aligned with the global feature map. The interaction is thus formulated as

$$Q_{\text{global}} = \text{MHCA}\left(Q + \text{PE}(A), \mathcal{F}_{\text{global}} + \text{PE}(X), \mathcal{F}_{\text{global}}\right) \quad (9)$$

where X represents the standard spatial grid coordinates of the global tokens. This step injects crucial macroscopic spatial priors, effectively circumventing the contextual limitations inherent to disconnected patch inference.

Local Microinteraction: Once imbued with global awareness, the global queries Q_{global} are dynamically routed and partitioned into subsets $\{Q^{(k)}\}_{k=1}^{K}$ corresponding to their assigned spatial patches. To perform precise microlevel feature extraction, the anchor boxes of each query subset must be projected from the global normalized coordinate system to the patch-local coordinate system. For a local patch $\mathcal{P}_k$ defined by its pixel boundaries (x_1, y_1, x_2, y_2) within an original image of width W and height H, the local continuous center coordinate cx_{local} is computed from the global center cx as

$$cx_{\text{local}} = \frac{cx \cdot W - x_1}{x_2 - x_1} \quad (10)$$

and similarly for cy_{local}. Alongside the proportional scaling of width and height, this transformation yields the localized anchor boxes $A_{\text{local}}^{(k)}$. Subsequently, a multiscale deformable attention (MSDA) [14] mechanism is applied independently within each patch group. This restricts the dense key–value interactions between the grouped queries $Q^{(k)}$ and their corresponding high-resolution local tokens $\mathcal{F}_{\text{local}}^{(k)}$

$$\widetilde{Q}^{(k)} = \text{MSDA}\left(Q^{(k)}, A_{\text{local}}^{(k)}, \mathcal{F}_{\text{local}}^{(k)}\right). \quad (11)$$

After the localized interaction, the refined query subsets $\{\widetilde{Q}^{(k)}\}_{k=1}^{K}$ are scattered back to reconstruct the unified query tensor, which is then processed by a feed-forward network. By decoupling the interactions into global and local hierarchical stages, the model achieves simultaneous perception of both macroscopic scene context and microscopic object details, ensuring highly precise bounding box refinement.

D. Optimization Objectives

The UHR-DETR is trained in an end-to-end manner. The overall loss function is formulated as a linear combination of the gain map generation losses and the set prediction losses

$$\mathcal{L}_{total} = \lambda_{map}\mathcal{L}_{map} + \lambda_{LPM}\mathcal{L}_{LPM} + \mathcal{L}_{det} \quad (12)$$

where $\mathcal{L}_{map}$ denotes the distribution focal loss for the gain map, and $\mathcal{L}_{LPM}$ represents the local peak margin loss. The detection loss $\mathcal{L}_{det}$ is computed based on the standard bipartite matching protocol utilizing the Hungarian algorithm, comprising varifocal loss for classification alongside L_1 and GIoU losses for bounding box regression.

IV. Experiments

A. Datasets

We conduct extensive experiments on two challenging large-scale datasets featuring massive image resolutions and densely packed small objects: STAR [3] and SODA-A [13].

1) STAR, originally proposed for scene graph generation, provides a challenging benchmark for object detection in satellite imagery. The dataset features high spatial resolutions (0.15–1 m), with original image dimensions spanning from 512 × 768 to 27 860 × 31 096 pixels. Overall, it contains over 210 000 instances annotated across 48 fine-grained categories.
2) SODA-A is a large-scale benchmark exclusively tailored for multicategory small object detection in aerial scenarios. It comprises 2513 high-resolution aerial images encompassing over 872 000 exhaustively annotated instances across nine distinct classes.
3) *Data Preprocessing:* To construct standardized evaluation benchmarks for UHR small object detection, we apply specific preprocessing strategies to both datasets. For STAR, we first select the eight categories exhibiting the smallest average object sizes across the entire dataset: *car*, *boat*, *tank*, *truck*, *boarding_bridge*, *crane*, *airplane*, and *lattice_tower*, unifying all annotations into the standard (x, y, w, h) representation. To formulate a specialized subset, we filter the dataset to retain only images that contain the instances of these selected categories and possess an original resolution exceeding 6 K. Finally, we employ a sliding window and zero-padding strategy to standardize the retained images to a fixed resolution of 8192 × 8192 pixels [5]. For SODA-A, we retain all original object categories. To obtain UHR-scale inputs consistent with our evaluation setting, we upsample the original images by a factor of 2. Since the long edge of a typical SODA-A image is approximately 4800 pixels, this operation increases it to approximately 9600 pixels. We then apply zero-padding along the shorter dimension to standardize all images to a fixed resolution of 9600 × 9600 pixels.

B. Evaluation Metrics

To evaluate detection performance, we report the mAP alongside AP_S, AP_M, and AP_L. All precision metrics are

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TABLE II

DETECTION RESULTS ON THE STAR DATASET. THE SUFFIX "@NK" DENOTES GLOBAL RESIZING OF THE INPUT IMAGE FOR TRAINING AND INFERENCE, WITH EACH SPATIAL DIMENSION CORRESPONDING TO n TIMES 1024 PIXELS. $*$ INDICATES TRAINING FROM SCRATCH FOR 36 EPOCHS FOLLOWING [34]

ID	Method	Backbone	Car	Boat	Tank	Truck	BB	Crane	Airplane	LT	mAP	AP_S	AP_M	AP_L	Latency (s)	Mem. (GB)
Exhaustive Sliding-Window Baselines (SW):																
#1	Faster-RCNN [41]	ResNet-50	5.3	11.6	39.0	10.7	51.0	17.6	80.4	50.9	33.3	14.8	34.6	38.8	4.839	4.85
#2	FCOS [19]	ResNet-50	4.6	6.5	35.4	7.3	50.2	14.0	79.0	38.6	29.5	12.5	31.1	36.6	3.384	2.27
#3	YOLOX-x [42]	CSPDarknet	5.1	1.6	25.3	7.7	18.4	7.3	72.8	25.5	20.4	7.8	21.7	27.2	5.244	3.62
#4	RT-DETR [12]	ResNet-50	4.7	10.9	30.9	7.0	62.3	16.2	80.5	44.0	32.1	14.0	33.1	37.7	3.068	2.32
Global Downsampling Approaches:																
#5	RT-DETR@1K [12]	ResNet-50	1.4	1.8	15.8	8.9	6.8	14.3	39.2	21.8	13.8	0.6	9.5	31.4	0.036	0.46
#6	Fovea@1K [6]	ResNet-50	0.0	0.0	12.6	4.3	3.3	8.2	35.2	19.3	10.3	0.2	6.4	24.8	1.560	1.97
#7	FCOS@1K [19]	ResNet-50	0.0	0.1	5.4	1.7	0.7	6.2	20.3	7.8	5.3	0.0	2.5	14.0	0.032	0.42
#8	FCOS@2K [19]	ResNet-50	2.7	0.2	24.4	5.2	7.4	10.5	56.7	21.8	16.1	0.9	14.7	29.7	0.100	1.01
#9	FCOS@3K [19]	ResNet-50	6.7	0.9	30.7	3.9	15.4	10.5	70.2	35.2	21.7	3.7	22.1	32.6	0.244	2.11
Micro-level Dynamic Routing & SW:																
#10	SparseFormer* [34]	SparseNet	4.9	0.6	25.2	5.8	44.8	12.7	74.2	17.3	23.2	7.6	23.8	30.4	4.146	2.51
#11	CEASC [36]	ResNet-50	5.6	8.5	31.8	10.4	51.5	16.2	78.2	43.1	30.7	14.7	31.4	36.7	3.820	5.00
Macro-level Patch Selection:																
#12	SPDet [10]	CSPDarknet	1.1	0.1	22.9	1.9	15.6	4.3	62.2	16.6	15.6	0.8	15.1	26.9	2.580	0.75
#13	GigaDet [32]	ResNet-50	1.1	1.2	20.8	3.7	23.5	1.3	50.7	11.5	14.2	4.6	15.2	17.8	0.345	3.12
#14	UHR-DETR (ours)	ResNet-50	14.5	10.4	34.6	33.9	52.4	20.2	70.1	43.2	34.9	15.0	36.4	40.5	0.357	3.13

calculated at a fixed intersection over union (IoU) threshold of 0.5. Additionally, we provide the average precision for each individual category for fine-grained analysis. To assess computational efficiency, we measure latency, defined as the average inference time per UHR image, and report the GPU memory footprint (Mem.) during inference for reference.

C. Implementation Details

We implement UHR-DETR using MMDetection [43] with a batch size of 1. All inference latency and peak inference memory measurements are conducted on an NVIDIA RTX 3090 GPU. Baseline configurations exceeding its 24-GB memory capacity during training are trained on a 48-GB GPU, while their inference measurements remain conducted on the RTX 3090. The backbone for extracting local features defaults to ResNet-50 [37], while a lightweight ResNet-18 [37] is utilized for extracting global features. Both backbones are initialized with ImageNet [44] pretrained weights. During the patch selection phase, the local patch size is set to 512 $\times$ 512 pixels. The number of selected patches K defaults to 40 for STAR, while it is set to 80 for SODA-A to accommodate the more spatially dispersed object distribution in the latter. These budgets are selected by balancing marginal coverage gains, accuracy, and latency. The discrete bin limit M is set to 6 based on dataset statistics, as over 99% of subregions exhibit a ground-truth coverage below 36 (M^2). For the decoupled decoder, the number of dynamically initialized object queries is bounded between 300 and 3000. The entire network is trained end-to-end for 12 epochs using the AdamW [45] optimizer with an initial learning rate of 1×10^{-4} and a weight decay of 1×10^{-4}. To preserve pretrained representations, the learning rate for both backbones is scaled by a factor of 0.1. We apply a linear warmup strategy for the first 500 iterations and decay the overall learning rate by a factor of 10 at the 8th and 11th epochs. Finally, gradient clipping with a maximum norm of 0.1 is adopted to stabilize the training process.

D. Main Results

To demonstrate the superiority of UHR-DETR, we conduct comprehensive comparisons with various state-of-the-art detectors on the STAR and SODA-A benchmarks. As detailed in Tables II and III, the competing methods are systematically categorized into four paradigms: exhaustive sliding-window baselines, global downsampling approaches, microlevel dynamic routing (with sliding windows), and macrolevel patch selection.

1) Results on the STAR Dataset: As presented in Table II, traditional sliding-window baselines achieve considerable accuracy but incur prohibitive computational costs. Uniform downsampling (#5) severely degrades small object detection. Although nonuniform downsampling (#6) attempts to preserve fine-grained features, it still struggles to perceive small objects in UHR images, even resulting in performance regression. Additional comparisons with FCOS at different input resolutions (#7–#9) show that increasing the resolution improves mAP from 5.3% at 1 K to 21.7% at 3 K but incurs rapidly

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TABLE III

DETECTION RESULTS ON THE SODA-A DATASET. THE SUFFIX "@NK" DENOTES GLOBAL RESIZING OF THE INPUT IMAGE FOR TRAINING AND INFERENCE, WITH EACH SPATIAL DIMENSION CORRESPONDING TO n TIMES 1024 PIXELS. ‡ INDICATES THAT THE PEAK TRAINING MEMORY EXCEEDS 48 GB. ∗ INDICATES TRAINING FROM SCRATCH FOR 36 EPOCHS FOLLOWING [34]

ID	Method	Backbone	AI	HE	SV	LV	SH	CO	ST	SP	WI	mAP	AP_S	AP_M	AP_L	Latency (s)	Mem. (GB)
Exhaustive Sliding-Window Baselines (SW):																	
#1	RT-DETR [12]	ResNet-50	87.9	54.5	50.0	41.4	69.1	39.7	69.4	71.9	49.5	59.3	29.5	67.2	48.7	4.370	2.33
#2	FCOS [19]	ResNet-50	82.8	31.6	50.2	32.9	65.1	34.1	65.5	70.0	53.9	54.0	26.3	61.4	43.2	4.754	2.27
Global Downsampling Approaches:																	
#3	RT-DETR@1K [12]	ResNet-50	29.0	23.4	2.3	4.8	8.1	19.0	11.1	48.0	19.0	18.3	1.2	21.0	34.3	0.036	0.46
#4	Fovea@1K [6]	ResNet-50	5.0	0.0	0.0	2.5	1.0	1.0	3.5	2.8	1.5	1.9	0.3	2.0	4.6	2.100	2.53
#5	FCOS@1K [19]	ResNet-50	6.8	0.6	0.5	0.3	2.3	6.8	2.2	17.6	4.7	4.6	0.4	4.9	15.4	0.032	0.42
#6	FCOS@2K [19]	ResNet-50	53.6	20.6	7.5	8.9	17.8	19.3	31.8	64.1	31.0	28.3	3.3	34.4	47.9	0.100	1.01
#7	FCOS@3K‡ [19]	ResNet-50	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Micro-level Dynamic Routing & SW:																	
#8	SparseFormer∗ [34]	SparseNet	80.0	62.6	47.1	37.3	61.4	46.8	63.3	83.1	40.6	58.0	24.0	68.0	53.6	4.802	2.51
#9	CEASC [36]	ResNet-50	81.6	61.4	45.7	37.6	63.6	43.0	66.5	80.2	56.1	59.5	25.7	69.7	48.3	4.467	5.38
Macro-level Patch Selection:																	
#10	SPDet [10]	CSPDarknet	64.5	8.2	31.9	20.9	42.8	38.8	55.1	70.8	18.0	39.0	13.1	48.5	39.3	1.957	0.75
#11	GigaDet [32]	ResNet-50	68.8	16.1	45.9	10.5	66.5	31.2	45.0	21.4	13.6	35.5	23.5	40.2	25.6	0.588	6.72
#12	UHR-DETR (ours)	ResNet-50	83.8	53.4	45.9	20.8	56.8	36.3	69.4	51.4	61.9	53.3	30.9	59.4	40.8	0.647	5.57

growing training costs. In particular, the peak training memory reaches 18.3 GB at 2 K and 42.7 GB at 3 K, while the detection accuracy remains substantially lower than that of UHR-DETR. The existing dynamic routing and patch selection approaches (#10–#13) either retain the high latency of sliding-window inference or achieve efficiency at the expense of detection accuracy. In contrast, our method (#14) reduces the inference latency to 0.357 s through the sparse extraction of fine-grained features, achieving nearly a 9× speedup compared to the sliding-window baselines. Furthermore, operating within a streamlined, fully end-to-end inference framework, our approach boosts the overall mAP to 34.9% and the small object accuracy (AP_S) to 15.0%. Particularly, for typical small objects, such as *car*, our method achieves a precision of 14.5%, significantly outperforming all other baselines.

2) Results on the SODA-A Dataset: To further validate the effectiveness of our method, we evaluate UHR-DETR on the SODA-A dataset under simulated 9600 × 9600 conditions. As shown in Table III, compared to sliding-window methods (#1 and #2), our approach (#12) achieves a competitive mAP of 53.3% while delivering a significantly faster inference latency of 0.647 s. For the globally resized FCOS baselines (#5–#7), FCOS@2 K requires 38.6 GB of peak training memory, while FCOS@3 K cannot be trained within 48 GB due to the substantially larger number of objects per image than in STAR. Simultaneously, our method yields the highest small object accuracy (AP_S of 30.9%) among all evaluated baselines. Even when facing more challenging conditions in this simulated scenario, our method successfully suppresses background redundancy. This indicates that the proposed sparse perception paradigm remains robust across varying scene complexities.

However, despite achieving strong performance, the margin of improvement on SODA-A is smaller than that on the STAR dataset. This is primarily because SODA-A is constructed via uniform upsampling to obtain UHR images, resulting in a more homogeneous object distribution. In contrast, the STAR consists of naturally acquired UHR imagery with more realistic observation distributions. Consequently, this difference poses greater challenges for efficient feature extraction in the Encoder. We further discuss this issue in detail in Section V.

E. Ablations

1) Effectiveness of Core Components: As detailed in Table IV(a), our proposed components (ISSGA, LPM, and GCA) synergistically drive a substantial performance leap, ultimately boosting the baseline mAP from 27.4% to 34.9%. The baseline model, which employs rigid patch selection (NMS, similar to [32]), yields 27.4% mAP. Replacing this rigid strategy with ISSGA brings a 2.1% mAP improvement (29.5%) by effectively mitigating the truncation of dense object clusters. The subsequent addition of the LPM objective triggers a

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TABLE IV

ABLATIONS. ISSGA, LPM, AND GCA DENOTE THE ISSGA, LOCAL PEAK MARGIN LOSS, AND GCA IN THE DECODER, RESPECTIVELY. GRAY ROW MEANS THE DEFAULT SETTING. (A) ABLATION STUDY ON THE CORE COMPONENTS OF OUR METHOD. (B) CROSS-EVALUATION OF THE NUMBER OF SELECTED PATCHES DURING TRAINING (K_{TRAIN}) AND INFERENCE (K_{INFER}). THE EVALUATION METRIC IS MAP. (C) SENSITIVITY ANALYSIS OF THE HYPERPARAMETER MARGIN Δ IN THE LOCAL PEAK MARGIN LOSS $\mathcal{L}_{LPM}$. (D) IMPACT OF THE LOCAL PATCH SIZE ON DETECTION PERFORMANCE AND LATENCY

(a)

Components			Metrics (%)			
ISSGA	LPM	GCA	mAP	AP_S	AP_M	AP_L
			27.4	11.1	29.2	32.8
		✓	28.3	11.9	30.0	32.8
✓			29.5	12.3	31.3	33.2
✓	✓		33.8	12.9	35.4	40.2
✓	✓	✓	34.9	15.0	36.4	40.5

(b)

K_{train}	Inference Patches (K_{infer})				
	20	30	40	50	60
20	29.9	31.4	31.8	31.8	31.7
30	30.6	32.3	33.0	32.9	33.0
40	31.7	33.7	34.9	35.3	35.6
Latency (ms)	238	298	357	420	479

(c)

Margin (Δ)	mAP	AP_S	AP_M	AP_L
0.00	33.6	16.2	35.3	39.6
0.05	34.9	15.0	36.4	40.5
0.10	33.2	13.2	34.6	40.2

(d)

Patch Size	K	mAP	AP_S	AP_M	AP_L	Latency (ms)
256×256	160	31.7	13.8	34.1	36.3	416
512×512	40	34.9	15.0	36.4	40.5	357
1024×1024	10	30.1	14.7	31.2	33.4	353

remarkable 4.3% mAP surge (reaching 33.8%). This highlights a crucial synergy: LPM explicitly sharpens local peaks in the gain map, providing optimal spatial guidance for the ISSGA routing. Finally, integrating GCA elevates the overall mAP to 34.9% and notably achieves a striking improvement for small objects (AP_S surges from 12.9% to 15.0%). This confirms that injecting macroscopic context into local decoding is essential for recognizing small objects with weak isolated features.

2) Impact of Patch Quantity (K): As shown in Table IV(b), our method demonstrates remarkable robustness to variations in patch budgets, enabling elastic, training-free adjustment of K_{infer} during deployment. Across training patch budgets, increasing K_{train} consistently raises the performance upper bound, as a larger budget exposes the network to more diverse object instances. Across inference patch budgets, the model is relatively insensitive to variations in K_{infer}. Increasing K_{infer} steadily improves mAP without requiring strict alignment with K_{train}, though it naturally introduces a linear latency overhead. For instance, with $K_{\text{train}} = 40$, pushing K_{infer} from 40 to 60 yields a marginal mAP gain (34.9%–35.6%) but substantially increases latency (357–479 ms). Balancing accuracy and efficiency, we adopt $K_{\text{train}} = 40$ and $K_{\text{infer}} = 40$ as our default configuration.

3) Sensitivity of the Margin Hyperparameter (Δ): As detailed in Table IV(c), $\mathcal{L}_{LPM}$ achieves optimal performance at $\Delta = 0.05$. Setting $\Delta = 0.00$ ensures a basic local maximum but lacks a distinct numerical gap, yielding 33.6% mAP. Introducing a positive margin of 0.05 effectively sharpens the gain map, providing precise spatial guidance for subsequent patch selection. Conversely, an excessively large margin ($\Delta = 0.10$, dropping to 33.2% mAP) imposes overly strict penalties that disrupt the natural continuous distribution of spatial features. Consequently, we adopt $\Delta = 0.05$ as our default configuration.

TABLE V

DIFFERENT PATCH SELECTION STRATEGIES

Selection Strategy	mAP	AP_S	AP_M	AP_L
Rigid Suppression (NMS)	30.0	12.3	30.6	35.8
Soft-Subtraction (Gaussian)	33.4	14.1	35.2	40.1
Soft-Subtraction (Linear, Ours)	34.9	15.0	36.4	40.5

4) Impact of Local Patch Size: As shown in Table IV(d), the 512×512 resolution emerges as the optimal sweet spot, delivering the best balance between detection accuracy and inference speed. Small patches (256×256) fragment the scene and truncate objects, reducing mAP to 31.7%, while the sheer volume of crops spikes latency to 416 ms. Conversely, large patches (1024×1024) encompass vast uninformative backgrounds, diluting the sparse perception objective and degrading mAP to 30.1%. Consequently, the 512×512 configuration provides the ideal spatial context, achieving peak accuracy while maintaining highly competitive efficiency.

5) Effectiveness of Patch Selection Strategies: As detailed in Table V, our linear soft-subtraction strategy significantly outperforms alternative mechanisms, achieving the optimal performance of 34.9% mAP. The baseline rigid suppression (NMS) limits the overall accuracy to a suboptimal 30.0% mAP. Transitioning to a Gaussian soft-subtraction kernel yields a substantial improvement (reaching 33.4% mAP), demonstrating the clear advantage of applying spatial overlap penalties. Ultimately, our linear decay kernel proves superior. Unlike the centralized smoothing effect of a Gaussian kernel, the linear penalty accurately models the uniform geometric nature of physical patch overlaps. This precisely aligns with our set

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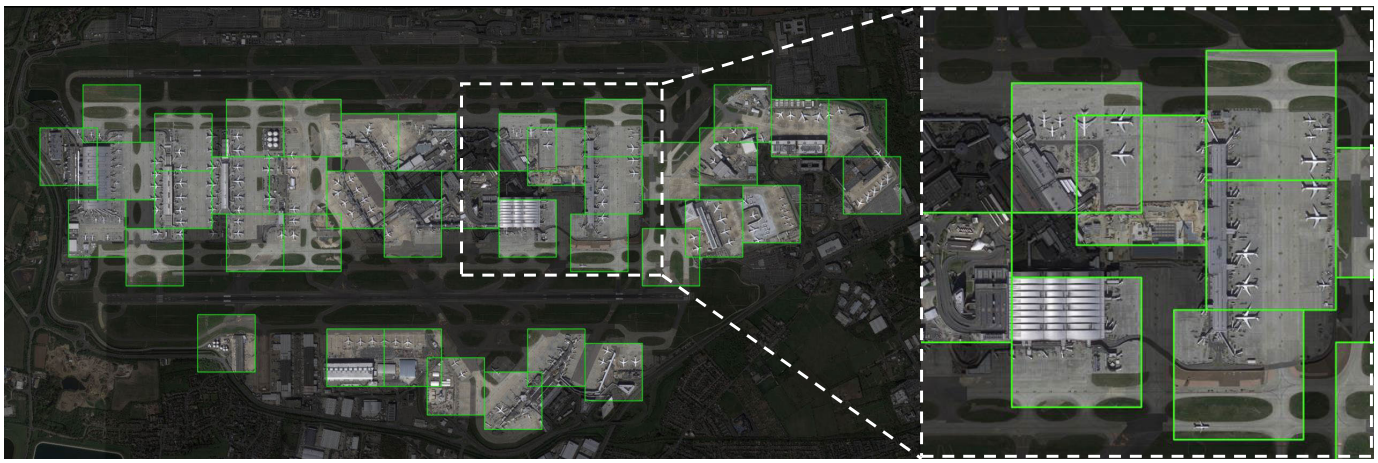


Fig. 3. Visualization of dynamic patch selection. Green boxes indicate the selected local patches. Our method efficiently concentrates computational resources on object-dense regions while filtering out vast uninformative backgrounds. A zoomed-in view is shown on the right.

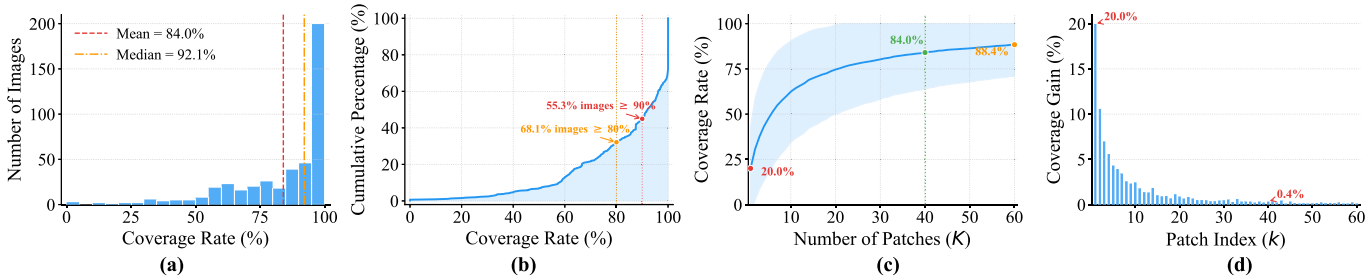


Fig. 4. Statistical analysis of object coverage rates. (a) Per-image coverage distribution. (b) Cumulative distribution curve (cdf) across the dataset. (c) Average coverage rate versus the patch budget (K). (d) Marginal coverage gain per patch iteration. These results collectively demonstrate that our dynamic allocation rapidly achieves high object coverage while efficiently managing spatial redundancy.

cover objective, maximizing object coverage while successfully avoiding redundant sampling.

F. Visualization

To validate the effectiveness of our sparse perception paradigm, we provide both qualitative and quantitative analyses. As visualized in Fig. 3, our method dynamically concentrates high-resolution patches on dense object clusters while successfully bypassing vast uninformative backgrounds, intuitively demonstrating its capability to minimize spatial redundancy. Across the STAR validation set, the statistical distributions in Fig. 4(a) and (b) corroborate this robustness, confirming that our strategy consistently maintains high overall object coverage. More importantly, as illustrated in Fig. 4(c) and (d), the coverage rate converges rapidly as the patch budget K increases, accompanied by a sharp decay in marginal coverage gain. This perfectly aligns with the diminishing returns property of our geometric set cover formulation, proving that our soft-subtraction greedy algorithm efficiently maximizes object perception within a limited computational budget.

Building upon this efficient patch extraction, Fig. 5 presents the final end-to-end detection results. As highlighted by the zoomed-in panels, our method accurately localizes densely packed small objects within the expansive UHR imagery. By seamlessly integrating macroscopic scene awareness into the decoding process, our method ensures robust and precise perception across continuous wide-area scenarios.

V. Discussion

In this section, we further clarify the design principles, empirical behavior, and inherent limitations of UHR-DETR by addressing three key questions.

1) Why use a gain map to model object distribution instead of coarse detection map or density maps?

The fundamental advantage of the gain map lies in shifting the global perception objective from precise object localization to regional value estimation. Conventional coarse detection [10], [29] and density mapping [30], [31] paradigms assume that the network can achieve fine-grained localization (e.g., predicting preliminary bounding boxes or precise object centers) during the heavily downsampled global stage. However, in massive UHR scenarios, small object features are severely obliterated after heavy downsampling, rendering such early-stage localization highly unreliable. In contrast, our gain map bypasses this bottleneck. Operating on macroscopic, low-resolution features, it only evaluates the relative information richness of each subregion (i.e., perceiving the expected quantity or value of objects within a local area) without demanding exact spatial coordinates. By aggregating coarse semantic responses to estimate regional value rather than individual object positions, this mechanism effectively tolerates weak initial signals, ensuring informative patches are reliably identified for subsequent high-resolution processing.

2) Why is macroscopic global context necessary when local patches already capture fine-grained object details?

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Fig. 5. Visualization of the final detection results in a massive UHR scenario.The zoomed-in panels on the right demonstrate our method's capability to accurately localize densely packed small objects. By effectively leveraging macroscopic context, the framework maintains robust perception across the expansive background.

In massive UHR scenes, the small objects often exhibit extremely weak isolated feature responses. Relying solely on localized crops deprives the network of the broader scene context needed to resolve semantic ambiguities. The global–local decoupled decoder solves this by using GCA to inject macroscopic structural cues into the localized decoding process. This mechanism fundamentally enhances the semantic representation and decisional certainty of isolated objects. As quantitatively demonstrated in Table IV(a), integrating this global context yields a 1.1% improvement in overall mAP. More critically, by effectively resolving semantic ambiguities, it drives a substantial 2.1% surge in small object precision (AP_S, from 12.9% to 15.0%).

3) What are the inherent limitations of the proposed UHR-DETR framework?

The primary limitation of UHR-DETR lies in its efficiency degradation when encountering homogeneously dense object distributions. While our dynamic patch allocation excels in typical UHR scenarios characterized by localized object clusters, its computational advantage diminishes in extreme failure cases, where objects are uniformly scattered across the entire canvas. In such uniformly dense distributions, the absence of spatial sparsity forces the network to allocate a massive number of patches (K_{infer}) everywhere to satisfy the set cover objective. Consequently, the overhead of the dynamic routing mechanism, combined with the sheer volume of processed regions, causes the inference latency to increase significantly, eventually approaching that of exhaustive sliding-window baselines. To mitigate this limitation, future work could explore content-adaptive patch allocation and multiresolution routing to dynamically adjust the number and resolution of processed patches according to regional difficulty, thereby preserving efficiency even when spatial sparsity is limited.

VI. Conclusion

In this article, we propose UHR-DETR, an end-to-end framework designed to resolve the fundamental conflict between preserving fine-grained features for small objects and managing the prohibitive computational cost of UHR imagery. By dynamically extracting information-rich regions and synchronizing them with macroscopic context, our method achieves highly efficient sparse perception without relying on exhaustive sliding windows. Extensive experiments on the STAR and SODA-A datasets demonstrate that our approach strikes an optimal balance between detection accuracy and efficiency, yielding nearly a 9× inference speedup over traditional baselines. Nevertheless, the UHR-DETR relies on the spatial sparsity of objects to achieve computational efficiency. In scenes with uniformly dense object distributions, a fixed patch budget may limit object coverage, while increasing the patch budget reduces its efficiency advantage over exhaustive sliding-window methods. Future work will focus on improving the adaptability of UHR object detection to more complex spatial distributions while maintaining high computational efficiency. We hope this work can provide a new paradigm and practical solutions for ultrawide-area remote sensing object perception.

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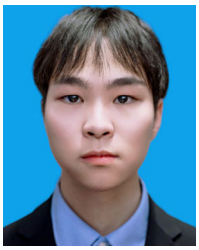
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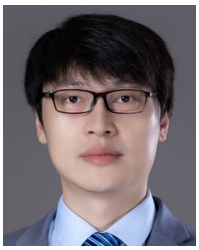
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